

Exp No: 5- ARDUINO BASED MANUAL ELECTRONIC COUNTER USING PROTEUS

Objectives:

1. To interface push buttons with the Arduino UNO for manual counting.
2. To display the counting values using a 7-segment display.
3. To understand the operation of a digital counter using Arduino and Proteus.

Program:

```
int seg[] = {1, 2, 3, 4, 5, 6, 7};
```

```
int button = 12;
```

```
int count = 0;
```

```
// a b c d e f g
```

```
byte digit[10][7] =
```

```
{
```

```
  {1,1,1,1,1,1,0}, // 0
```

```
  {0,1,1,0,0,0,0}, // 1
```

```
  {1,1,0,1,1,0,1}, // 2
```

```
  {1,1,1,1,0,0,1}, // 3
```

```
  {0,1,1,0,0,1,1}, // 4
```

```
  {1,0,1,1,0,1,1}, // 5
```

```
  {1,0,1,1,1,1,1}, // 6
```

```
  {1,1,1,0,0,0,0}, // 7
```

```
  {1,1,1,1,1,1,1}, // 8
```

```
  {1,1,1,1,0,1,1} // 9
```

```
};
```

```
void setup()
{
  for (int i = 0; i < 7; i++)
  {
    pinMode(seg[i], OUTPUT);
  }

  pinMode(button, INPUT_PULLUP);

  showNumber(0);
}

void loop()
{
  if (digitalRead(button) == LOW)
  {
    delay(50);

    if (digitalRead(button) == LOW)
    {
      count++;

      if (count > 9)
      {
        count = 0;
      }

      showNumber(count);
    }
  }
}
```

```
    // Wait until button is released
    while (digitalRead(button) == LOW)
    {
        delay(10);
    }
}
}
```

```
void showNumber(int n)
{
    // Turn OFF all segments first
    for (int i = 0; i < 7; i++)
    {
        digitalWrite(seg[i], LOW);
    }

    delay(20);

    // Display selected number
    for (int i = 0; i < 7; i++)
    {
        digitalWrite(seg[i], digit[n][i]);
    }
}
```

Output:

